Lab 6 Designing a study in the islands

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1. In this experiment, I will test islanders' IQ and the speed at which they complete the puzzle cube.

   The purpose of this experiment is to find out if there is a correlation between the Islanders' IQ and the speed of completing the puzzle cube.

   Independent Variable: the IQ of the islanders The islanders will be categorized into different IQ level groups, such as the high IQ group, the medium IQ group, and the low IQ group.

   Dependent Variable: speed of completing the puzzle cube.

   Motivation: The motivation for this experiment is to explore the potential relationship between the Islanders' IQ and their cognitive ability and learning speed. This could help to understand the impact of IQ on Islanders' ability to solve complex tasks such as the puzzle cube. This study could also provide interesting insights into the educational and cognitive development of Islanders.

2. I want to generalize the findings to the entire population of the three islands, focusing on individuals who have never played puzzle cubes before, regardless of gender.

3. "Is there a relationship between the IQ levels of islanders who have never tried puzzle cubes before and their performance in completing puzzle cubes?"

4. a) I will do a randomized controlled experiment with a repeated measures design.

   b) I will sort the 40 islanders into groups based on their IQ levels to create three groups: a high IQ group (120+), a medium IQ group (90-120), and a low IQ group (90-). Pure random assignment will be used to ensure that individuals within each IQ group are assigned to either the puzzle cube task or the control group.

   c) For the repeated measures design, all groups, including the control group, will complete the puzzle cube task. The control group will also complete a placebo task to control for any potential learning effects.

5. a) Group Selection: This threat potentially applies to the study. If there is a significant bias in the initial grouping of participants based on their IQ levels, it could impact the causality argument of the study. If the random assignment process is not effective in balancing the groups and results in one group having a large number of participants with a particular characteristic, it could introduce bias.

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b) Setting: The threat of setting does not apply to this study. The experiment can be controlled within a consistent setting, and the islanders' environment would not significantly affect the study's validity.

c) Timing: Timing is relevant to this study, as the time intervals between the tasks and the response measures need to be carefully considered. If the time intervals are too short or too long, it could affect the participants' performance or the effectiveness of the placebo task, potentially impacting the causality argument.

d) Test Familiarity: This threat applies to the study, especially with the cube puzzle task. If participants have previous experience with the cube puzzle or if they are more familiar with the task, it could confound the results. To mitigate this threat, I will only select islanders who have never played puzzle cubes before to complete this task.

6.

ID	Name	Village	Household_Num	Marital_Status	Time	Measurement
1	RENEE JONES	Colmar	50	Divorced	Pre	7.7
1	RENEE JONES	Colmar	50	Divorced	Post	7.5
2	DR ESTHER SOMMER	Hofn	10	Single	Pre	9.7
2	DR ESTHER SOMMER	Hofn	10	Single	Post	8.6
3	NAOKI SATO	Helvig	478	Single	Pre	8.6
3	NAOKI SATO	Helvig	478	Single	Post	7.3
4	JAKOB CARLSEN	READING	6	Single	Pre	7.6
4	JAKOB CARLSEN	READING	6	Single	Post	7.8
5	ORIEL ABEL	Pauma	19	Single	Pre	9
5	ORIEL ABEL	Pauma	19	Single	Post	7
6	CHAHEL DATTA	Maeva	468	Single	Pre	6.7

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6	CHAHEL DATTA	Maeva	468	Single	Post	7.2
7	MANA SHAH	Vaiku	325	Married	Pre	9.3
7	MANA SHAH	Vaiku	325	Married	Post	8
8	RANDALL BROWN	Shinobi	302	Single	Pre	8
8	RANDALL BROWN	Shinobi	302	Single	Post	7.3
9	EMMA EKLUND	Bjurholm	321	Married	Pre	7.5
9	EMMA EKLUND	Bjurholm	321	Married	Post	8.6
10	NAOKO ISHII	Nelson	344	Single	Pre	6.5
10	NAOKO ISHII	Nelson	344	Single	Post	7

ID	**Name**	**Village**	**Household_Num**	**Age**	**TimePC/min**	**IQ**
1	RENEE JONES	Colmar	63	54	6.7	124
2	DR ESTHER SOMMER	Hofn	10	60	7.2	117
3	NAOKI SATO	Helvig	478	12	9.3	92
4	JAKOB CARLSEN	READING	6	22	8	107
5	ORIEL ABEL	Pauma	19	39	8	100
6	CHAHEL DATTA	Maeva	468	29	7.3	114
7	MANA SHAH	Vaiku	325	40	7.5	112
8	RANDALL BROWN	Shinobi	302	13	/	71
9	EMMA EKLUND	Bjurholm	321	39	/	75
10	NAOKO ISHII	Nelson	344	37	7	134